

Victoria's public transport services are spatially and temporally unbalanced.

1. Introduction

This report focuses on the spatial and temporal differences in public transport services across different regions of Victoria, covering all Local Government Areas (LGAs). Primarily from the perspective of public transport service provision, the project uses GTFS timetable data, LGA boundary data, and monthly patronage supplementary data. This analysis focuses on the characteristics of different regions in terms of service density, coverage, operating hours, and service structure. The research objective is to identify the main spatial and temporal differences in public transport services across different regions of Victoria and to compare how these differences exhibit different patterns depending on mode of transport, temporal conditions, and region type.

The reason for choosing this topic is that primarily from my personal daily commuting experience. I frequently commute by bus and train in Victoria, which has given me a direct understanding of how public transport services differ across regions. In my daily commutes, I've noticed significant variations in frequency, service duration, and mode of transport options across different areas. Some regions offer more convenient transport connections, while others experience greater limitations during off-peak hours or on weekends. Based on the topic and the observations above, this report addresses the study through three research questions:

1. How do public transport service availability and spatial coverage vary across areas of Victoria by transport mode, time of day, and weekday versus weekend periods?
2. What patterns emerge when comparing stop density, service frequency, transport mode variety, and service hours across local areas in Victoria?
3. Do public transport systems across Victoria follow a single trade-off between service concentration and spatial reach, or do different structural patterns emerge across local areas?

2. Data Checking and Data Wrangling

2.1. Data sources

This report primarily uses three types of data. The first is the Victorian Public Transport GTFS Schedule data, which provides timetables and station information for different modes of transport and is the core data source for this report. The second is LGA boundary data, used to aggregate public transport data to the spatial unit of Local Government Areas (LGAs) to compare differences between different regions in Victoria. The third is Monthly average patronage by day type and by mode data, used as supplementary data. These three types of data support this study from the perspectives of service supply, spatial scope, and passenger flow, respectively. All the data sources we used are listed in the reference.

2.2. Data checking

Before checking the GTFS data, we first browsed the file structure within the GTFS compressed package to confirm its basic composition. After browsing the contents of the GTFS compressed package, we found that it contains several folders named with numerical codes. By reading the [GTFS Release Notes](#), we confirmed that these numerical codes correspond to different types of transportation services. Based on this correspondence, we then decompressed the package in R and renamed these folders to more easily understood transportation type names, facilitating the subsequent reading and organization of relevant data tables according to different transportation modes. LGA data is also provided in compressed file format, but its structure is relatively simple, so it only needs to be decompressed once. After decompression, the boundary files required for subsequent spatial analysis are obtained. Monthly average patronage data is provided directly as a CSV file and does not require additional decompression.

Due to the large volume of GTFS data, this report first categorizes the original public transportation data into three types: Train, Bus, and Tram. This can reduce the complexity of subsequent checks and comparisons. Based on this, data checks primarily focus on six core tables: stops, stop_times, trips, routes, calendar, and calendar_dates. For these six core tables across the three transportation modes, we first performed an NA value check on key attributes. The results (Figure 1) showed that none of

the key fields required for the current analysis contained missing values. Therefore, NA-based deletion was not necessary at this stage. To further ensure the reliability of subsequent analyses, we also performed a simple check of the main key-value relationships between the six core tables, and no unmatched records were found (Figure 2), indicating that these tables can be joined correctly in subsequent analyses.

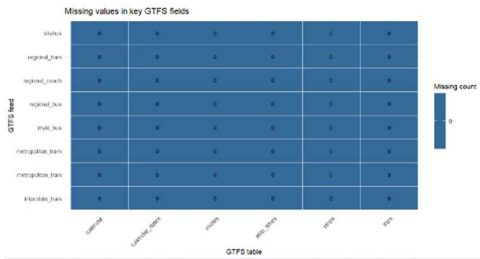


Figure 1, Missing values in key GTFS.

GTFS key matching check				
GTFS feed	Unmatched trip_id in stop_times & trips	Unmatched stop_id in stop_times & stops	Unmatched route_id in trips & routes	Unmatched service_id in trips & calendar
interstate_train	0	0	0	0
metropolitan_train	0	0	0	0
metropolitan_tram	0	0	0	0
myki_bus	0	0	0	0
regional_bus	0	0	0	0
regional_coach	0	0	0	0
regional_train	0	0	0	0
skybus	0	0	0	0

Figure 2, GTFS Key matching check

For LGA spatial data, the focus of data inspection is not on missing values, but on confirming that the boundary files can be read correctly and that there are no obvious spatial errors. Therefore, this report first uses R to draw a nationwide LGA boundary map for preliminary inspection to confirm that the layers are displayed normally, the boundary range is complete, and the data itself does not have any obvious anomalies (Figure 3).

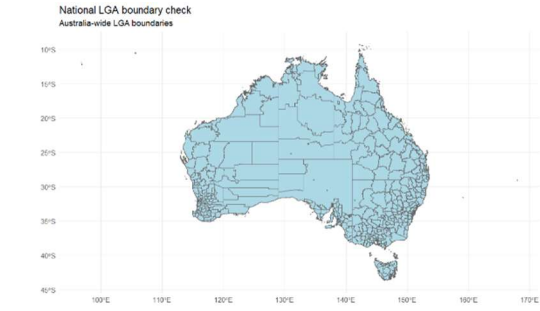


Figure 3, National LGA boundary check

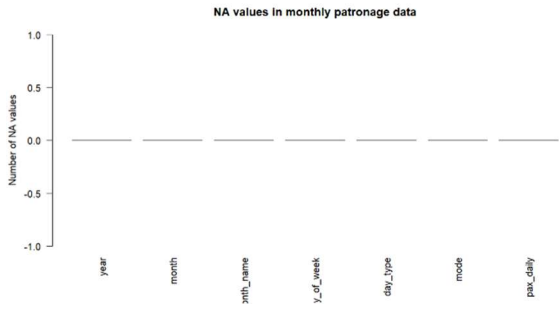


Figure 4, NA value check in monthly patronage data

For monthly average patronage data, the inspection method is relatively simple. Mainly involving statistical analysis of the NA values in each field. The results show (Figure 4) that this supplementary data does not contain any missing values that need to be addressed in the currently used fields.

Therefore, this supplementary data can be directly used for subsequent analysis.

2.3. Data Wrangling

Although the report structure indicates that data checking should precede data wrangling, but the two are not entirely separate. We completed some data processing simultaneously while checking the GTFS data. Due to the large volume of GTFS data and the fact that the original data was stored separately according to multiple transportation service types. We first further categorized these original types into three main categories: Train, Bus, and Tram. It's important to note that these three categories do not correspond to a single table, but are composed of multiple original transportation types. Therefore, the first step in the data processing was to merge the common core tables within each category. For example, multiple original types belonging to Train would first have their corresponding stops, stop_times, trips, and calendar tables merged. Bus and Tram were processed in the same way. The purpose of this was to organize the originally scattered GTFS data into a unified structure of three major transportation modes, laying the foundation for subsequent analysis.

After completing this initial data processing, we did not merge all the core tables into a single master table. Instead, we created targeted table connections based on the three research questions. For the first question, focusing on spatial coverage and temporal variation, we primarily combined tables related to stations, timetables, journeys, and service days. These tables will be used to construct the coverage and temporal patterns of different modes of transportation at the LGA level. For the second question, focusing on comparing station density, service frequency, mode diversity, and service duration. We further organized and summarized the relevant tables at the LGA level to form comparable indicators. For the third question, based on the previously compiled indicators, we combined key variables such as spatial coverage and service concentration to observe the structural relationships between public transportation systems in different regions. This approach avoids unnecessary large-scale merging and ensures that the data tables used for each research question are consistent with the analytical objectives.

3. Data Exploration

3.1. Question 1: Spatial and temporal differences in public transport services

To answer this question, this report analyses the data from two dimensions: spatial coverage and temporal variation. Spatially, a hierarchical color-coded map is used to compare the site coverage of each LGA, as maps can visually represent regional differences on a real-world geographic scale.

Temporally, heatmaps are used to compare service intensity across different modes of transport, time periods, and weekdays and weekends, as this format is better suited than maps for displaying differences between multi-category variables. Additionally, a small table is used to supplement the overall comparison results for weekdays and weekends.

In the spatial analysis, this report uses the number of stations in each LGA as an indicator of public transport coverage. As shown in Figure 5, there are significant differences in spatial coverage among different modes of transport in Victoria. Buses have the widest coverage, with stations observed in most LGAs, indicating their stronger role in regional connectivity. Trams are most concentrated, mainly located within the Melbourne metropolitan area. It is indicating their clear reliance on the city's core area. Trains fall in between buses and trams. Although their coverage is not as wide as buses, it is significantly more dispersed than trams and covers a larger area of Victoria's LGAs. The distribution near some state borders in the figure also suggests that the railway system not only serves intra-state connections but may also play a role in inter-state connections. Overall, Figure 5 illustrates that the spatial coverage of public transport in Victoria does not exhibit a single pattern but rather shows significant differences depending on the mode of transport.

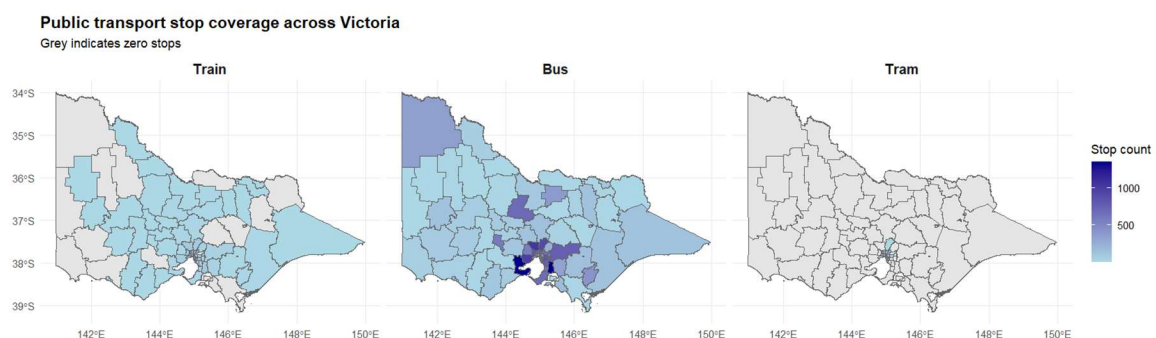


Figure 5, Public transport stop coverage across Victoria

Since this section involves three dimensions: mode of transport, time of day, and weekday/weekend. So, heatmaps are more suitable than maps or bar charts for illustrating the strength differences between different combinations. To compare service differences across different time periods, this paper divides departure times into three periods: morning peak (07:00–09:59), midday (10:00–14:59), and evening peak (15:00–19:59). As shown in Figure 6, passenger traffic for trains, buses, and trams increases over time on weekdays, peaking during the evening peak. While weekend peak traffic is concentrated around midday. Buses maintain the highest passenger traffic throughout all time periods. Overall, public transport in Victoria does not exhibit a single time pattern but varies depending on the mode of transport and the day of the week.

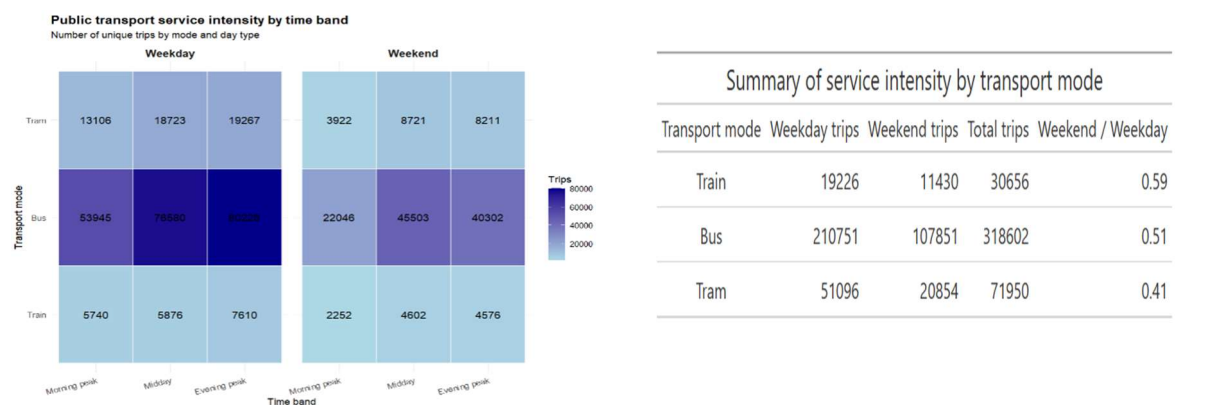


Figure 6, Public transport service intensity by time band

Table 7, Summary of service intensity by transport mode

Summary of service intensity by transport mode				
Transport mode	Weekday trips	Weekend trips	Total trips	Weekend / Weekday
Train	19226	11430	30656	0.59
Bus	210751	107851	318602	0.51
Tram	51096	20854	71950	0.41

Since this section requires a direct comparison of the overall service volume and the percentage change of different modes of transportation on weekdays and weekends, tables are more suitable than graphs for clearly presenting specific values. Therefore, Table 7 is used for supplementary explanation. Table 7 summarizes the overall service intensity of different modes of transport on weekdays and weekends. Trips represent the total number of independent trips for each mode of transport. And Weekend/Weekday represents the proportion of service retained on weekends relative to weekdays. So, the number of trips for all three modes of transport is lower on weekends than on weekdays. Buses maintain the highest service volume in both day types and are the most prevalent public transport mode in Victoria. Trains and Trams have relatively lower total service volumes, with Trams showing the most significant decrease on weekends (0.41), while Trains have the highest service retention rate on weekends (0.59). Overall, this table illustrates that different modes of

transport not only differ in total service volume but also in the magnitude of their changes between weekdays and weekends.

The three charts together show that different modes of transport in Victoria play different service roles in terms of space and time. Buses have the widest coverage and highest service volume, making them more suitable for providing extensive and continuous regional connections. Trams are mainly concentrated in urban core areas, serving more short-distance urban travel. Trains fall in between, serving both intra-state regional connections and supporting cross-regional connections to some extent. In terms of time, buses maintain high service volume on both weekdays and weekends, train service is relatively more stable on weekends, while tram service drops most significantly on weekends, indicating that it is more dependent on weekday urban activities and commuting demand. Overall, different modes of transport do not provide the same service under the same conditions, but rather correspond to different spatial ranges, travel distances, and time requirements, thus exhibiting different spatiotemporal distribution patterns.

3.2 Comparison of public transport characteristics of different LGAs

Unlike the first question, which focuses primarily on the distribution of public transport services. But the second question focuses more on the structure of public transport systems in different regions. Therefore, this section revolves around the second research question, comparing the differences in station density, mode diversity, and service duration among different LGAs in Victoria, and observing the main patterns presented by these indicators.

Stop density is defined as the ratio of the total number of public transport stops in each LGA to the area of the region, representing the density of stops per square kilometer. This metric is more suitable for comparing differences between LGAs of different areas than simply using the total number of stops. As shown in Figure 8, there are significant differences in station density across different LGAs in Victoria. The station density is higher in the Melbourne metropolitan area, while it is relatively lower in most other areas, indicating that public transport stations are mainly concentrated in more urbanized areas. Overall, station density reflects the significant differences in public transport infrastructure among the different LGAs.

Mode variety refers to the number of public transportation modes actually present in each LGA.

Modes of transportation are categorized into three types: Train, Bus, and Tram. Therefore, the value ranges from 0 to 3, with higher values indicating a richer transportation structure. As shown in Figure 9, mode variety exhibits a clear spatial stratification. Areas with a value of 3 are mainly concentrated near the Melbourne metropolitan area. Areas with a value of 2 are distributed not only on the outskirts of Melbourne but also along state borders, indicating that some border areas maintain certain transportation connections with other states. More remote LGAs have a value of only 1, indicating that they mainly rely on a single mode of transportation.

Service hours represent the time span between the earliest and latest departure records in each LGA, reflecting the duration of public transport service throughout the day, rather than the number of services themselves. As shown in Figure 10, there are also significant differences in service duration among different LGAs in Victoria. Areas with longer service durations are mainly concentrated in areas with stronger public transport activity, while some remote or regional LGAs have relatively shorter service durations. This indicates that different areas differ not only in the number of stations, service frequency, and diversity of transport modes, but also in the duration of public transport operation. These figures collectively reveal that public transport systems differ across the different LGAs (Local General Areas) in Victoria. The Melbourne metropolitan area and surrounding regions typically exhibit higher station density, mode diversity, and service duration, while more remote or regional LGAs generally show lower indicators. Meanwhile, some border areas demonstrate an advantage in mode diversity, suggesting their role in connecting other states or surrounding regions. Overall, the second research question indicates that different LGAs play different service roles in the public transport system.



Figure 8, Stop density across LGAs



Figure 9, Service frequency

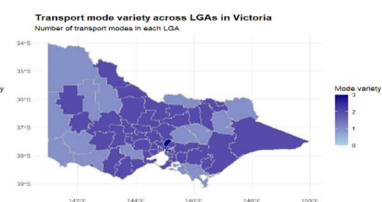


Figure 10, Transport mode variety

3.3 Structural Patterns of Victoria's Public Transport System

This section addresses the third research question: whether public transport systems in different LGAs of Victoria follow a single relationship: service concentration-spatial coverage or whether they exhibit different structural patterns. Unlike the first two questions, which focus on service distribution and system structure respectively, this section emphasizes the relationships between different indicators.

In terms of visualization, the reason for choosing a scatter plot is that we need to compare whether there is a trend between the two-core metrics. Also, scatter plots are better suited for observing whether variables exhibit linear relationships, whether outliers exist, and whether different regions will form different structural patterns. Each point in the figure represents an LGA. The horizontal axis represents the spatial coverage area. The vertical axis represents the service concentration. The color and the size of the point respectively supplement the diversity of transportation modes and the service duration.

As shown in Figure 11, the public transport systems of different LGAs in Victoria generally exhibit a positive correlation, meaning that the more stations a LGA has, the more independent service trips it typically offers. However, this relationship is not entirely consistent. The lower left corner shows a large cluster of points, indicating that many LGAs have both fewer stations and lower service volumes, resulting in a smaller overall public transport scale. A few points in the upper middle part of the figure show that some LGAs, while not having the most stations, have significantly higher service concentration, reflecting strong service clustering characteristics. Conversely, some points on the right side, although having more stations, have not seen a correspondingly significant increase in service volume, indicating that these areas prioritize broad coverage over high-frequency service.

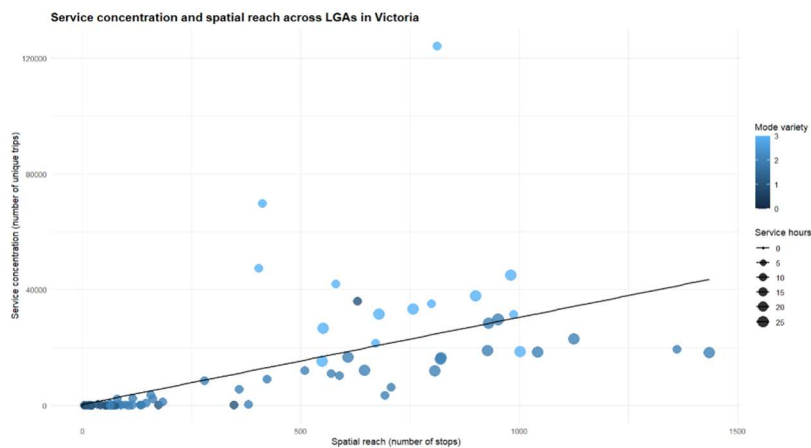


Figure 11

4. Conclusion

Train, Bus, and Tram exhibit significant differences in spatial coverage, temporal distribution, and system structure. Buses have the widest coverage, maintaining peak service volume on both weekdays and weekends, making them Victoria's primary mode of public transport. Trams are most concentrated, primarily located in the Melbourne metropolitan area, with the most significant service drop on weekends. Trains fall somewhere in between, with less coverage than Buses but more dispersed than Trams, and a relatively higher level of weekend service retention. At the regional level, the Melbourne metropolitan area and surrounding regions typically have higher station density, more diverse transport modes, and longer service durations, while more remote or regional LGAs generally have lower indicators, indicating that public transport systems differ in scale and structure across different regions. Furthermore, while public transport systems across Victoria generally show a trend of "more stations, more service," this relationship is not singular, and different LGAs still exhibit different structural combinations.

Overall, this report demonstrates that public transport services in Victoria exhibit significant imbalances across different modes of transport, regions, and time conditions. Meanwhile, this study mainly answers where the differences are manifested and what patterns they present, but does not further explain the reasons for these differences. Future studies could combine variables such as population, land use, and commuting demand to conduct a more in-depth analysis of the underlying mechanisms.

5. Reflection

Through this project, I've gained a clearer understanding that data visualization is not just about displaying results, but also about helping to understand the structure of a problem. Different problems require different types of graphics: maps are better for showing spatial distribution, heatmaps are better for comparing time patterns, and scatter plots are better for observing the relationships between variables. One mistake I made was that I didn't pay enough attention to what each chart and table was answering when I first started writing the report, which led to repetitive explanations and unclear chart functions in some paragraphs. In the future, I will determine the specific function of each chart earlier in my writing before developing the textual analysis.

References:

1. GTFS Schedule (TXT/CSV File) for question 1, 2 and 3.
<https://opendata.transport.vic.gov.au/dataset/gtfs-schedule>
2. Australian Bureau of Statistics. (2021, July 20). *Digital boundary files*. Local Government Areas - 2025 – Shapefile.
<https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/access-and-downloads/digital-boundary-files>

Appendix:

1. Link For R Script and Figures:
https://drive.google.com/drive/folders/16SzEN2wSjLdJBa0MzlZc6z87ElmZkL_z?usp=sharing

Acknowledging the use of Google Translate

I used Google Translate tool to help improve the accuracy of sentence through out this proposal. I modified the outputs by revising the phrasing, correcting expressions, and adjusting the sentences to better reflect my own meaning.